

Arghamitra Talukder

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EDUCATION

2023 - 2028	PhD in Computer Science at Columbia University <i>NSF CSGrad4US Fellow</i>	New York, NY GPA: 3.98/4.0
2018 - 2021	BS in Electrical Engineering at Texas A&M University <i>Engineering Honors, Magna Cum Laude</i>	College Station, TX GPA: 3.78/4.0

PROJECTS

Predicting Exon Inclusion Levels via Contrastive Sequence Models

Developed a deep learning model to analyze gene regulation with a contrastive learning framework. This model identifies conserved functional elements by mapping evolutionarily related DNA sequences into a shared latent space, providing novel insights into gene expression.

Bayesian hierarchical isoform quantification with hybrid sequencing

Developed JOLI, a novel hierarchical model that enhances RNA transcript quantification by jointly integrating short-read (SR) and long-read (LR) sequencing data. The model utilizes a multi-sample learning framework to improve estimation accuracy and reproducibility, particularly for low-abundance isoforms.

Modeling cell plasticity with integrated single-cell data

Designed a statistical model to quantify cellular plasticity by integrating single-cell lineage and gene expression data. This approach utilizes a continuous annotation distribution methodology to statistically detect and score plastic cell states.

PUBLICATIONS

Talukder, Arghamitra, Nicholas Keung, et al. (2025). “CLADES — Contrastive Learning Augmented Differential Splicing with Orthologous Positive Pairs”. In: *RECOMB (Under review)*.

Talukder, Arghamitra, Shree Thavarekere, et al. (2025). “Multi-sample, multi-platform isoform quantification using empirical Bayes”. In: *RECOMB-Seq (Short-talk)*. URL: <https://www.biorxiv.org/content/10.1101/2025.02.08.637184v2.abstract>.

Talukder, Arghamitra and Itsik Pe’er (2024). “Computational Method to Detect Cancerous Plasticity”. In: *IICD Workshop on Single-Cell Data Integration & OT, CRA-WP Grad Cohort for IDEALS (Short-talk)*.

Talukder, Arghamitra, Rujie Yin, et al. (2022). “Does inter-protein contact prediction benefit from multi-modal data and auxiliary tasks?”. In: *MLSB, NeurIPS*. URL: https://www.mlsb.io/papers_2022/Does_Inter_Protein_Contact_Prediction_Benefit_from_Multi_Modal_Data_and_Auxiliary_Tasks.pdf.

Ibrahim, Bassem, **Talukder, Arghamitra**, and Roozbeh Jafari (2020). “Multi-source multi-frequency bio-impedance measurement method for localized pulse wave monitoring”. In: *EMBC, IEEE*. URL: <https://ieeexplore.ieee.org/abstract/document/9176495>.

WORK EXPERIENCE

Product & Test Engineer at Texas Instruments, Dallas June 2021 - Aug 2023

- Responsible for TI customized Ultra-High Voltage test technology with 70% cost efficiency.
- Created a product specific test program converting from legacy tester to the latest technology reducing 40% of the test time.

Test & Validation Intern at Texas Instruments, Dallas June 2020 - Aug 2020

- Enabled traceability with the accuracy of 99.9% by device ID recovery obtained using OCR technology.
- Automation of data acquisition and validation by database access and scripting using SQL and PERL.

FELLOWSHIPS

NIH T15 Training Grant	Sep 2025 - Aug 2026
CISE CSGrad4US Fellowship, CRA, NSF	Sep 2023 - Aug 2027
NeurIPS Travel Award	Dec 2022

TEACHING ASSISTANT

Machine Learning for Functional Genomics	Sep 2025 - Dec 2025
Computational Genomics	Jan 2025 - May 2025
Experimental Physics Lab II-Mechanics	Jan 2019 - Dec 2019
Foundation of Engineering in Python and LabVIEW	Sep 2018 - Dec 2018

VOLUNTEERING

Columbia pre-submission application reviewer	Nov 2025, 2024
Conference volunteer at MLCB	Sep 2025
Panelist at the critical thinking in STEM career panel, NYGC	June 2024

MENTORING

Nicholas Keung, MS, Columbia CS	Jan 2025 - Ongoing
Tanya Mehta, MS, Columbia CS	Sep 2025 - Ongoing
Abhinav Goel, Freshman, Columbia CS	Sep 2025 - Ongoing
Shree Thavarekere, MS, Columbia CS	Jan 2024 - Jan 2025
Claire Dawson, MEng, Columbia CS	Jan 2024 - May 2024